

C9555: A practical guide to simulations

Week 7

Denys Biriukov – Oct. 30th, 2025

Get ready for practical session

- Open new terminal `Ctrl + Alt + T`
- Course materials at: https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555
- Any technical issues/problems/questions? Checklist:
  - Access to Wolf.
  - Proper infinity and uv environments.

More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let's chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

Get ready for practical session

- A bit of things to be done:
  - `mkdir C9555`
  - `mv ~/survival_guide_simulations_c9555/week04/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

Get ready for practical session

- `git clone --depth 1 https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week07/`

- `bash ../infinity/infinity-temp.sh`
- `bash ../uv/uv-activate.sh`
- `bash ../uv/jupyter-start.sh`

What is today?

- Week 07:
  Adding salt to our water
- We have water!
  - The main solvent of life
- We all add salt in water at some point.
- Today we will do it in MD simulations.
- Will it behave realistically?

TEACHING WEEK	DATE	LECTURER	TOPIC
01	18.09	WM	Simple MD to Describe Materials Part I
02	25.09	WM	Simple MD to Describe Materials Part II
03	02.10	IK	Visualization
04	09.10	DB	How to run MD simulations?
05	16.10		**BREAK**
06	23.10	IK	Sampling in Molecular Dynamics
07	30.10	DB	Adding salt to our water
08	06.11	DB	
09	13.11	DB	
10	20.11	IK	
11	27.11	IK	
12	04.12	WM	
13	11.12	WM	

Today program

Lecture

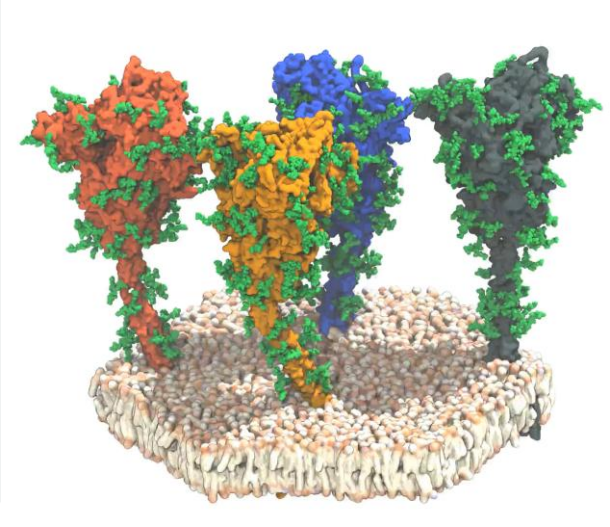
- Force fields: what, how, and when.
- Bonded and nonbonded interactions: origin and implementation.
- Ewald summation.

Practical session

- Preparing solvent–solute systems: ions in water.
- Solute–solute and solute–solvent interaction-driven phenomena and their analysis.

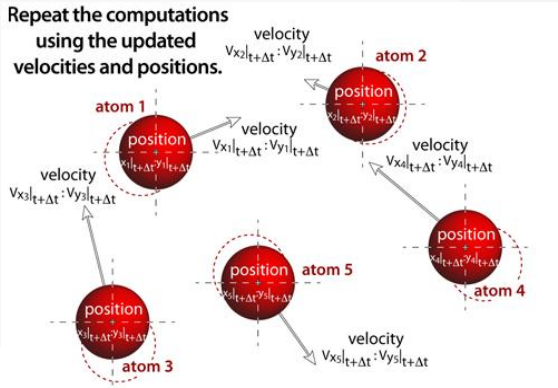
Quick recall of the previous week

- Molecular dynamics simulations are based on physical laws and concepts.
- Atoms are represented as billiard balls that move according to Newton's equations of motion.
  - Born–Oppenheimer approximation.
- Relation of simulation trajectories to thermodynamics.
- Time and ensembles.



Spike protein of SARS-CoV-2

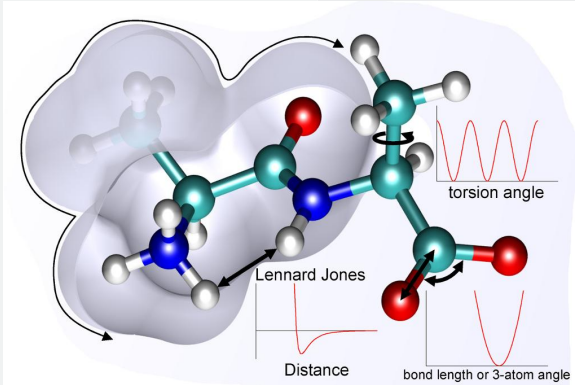
TUROŇOVÁ, SIKORA, SCHÜRMANN et al.
MAX PLANCK INSTITUTE OF BIOPHYSICS
PAUL-EHRLICH-INSTITUT



http://atomsinmotion.com/book/chapter5/md

Force field: basic information

- The force field (FF, V) determines the force that an atom feels.
- The force field is “invariable” during a simulation.
- The force fields are usually additive/modular.
- The force field is the collection of functions and its parameters.
- There are many force fields for biosimulations.
  - Different in the functional forms, which reflect certain philosophy.
  - The same functional form but simply different parameters.
- Is there the ultimate and the best of the best force field?
  - No.

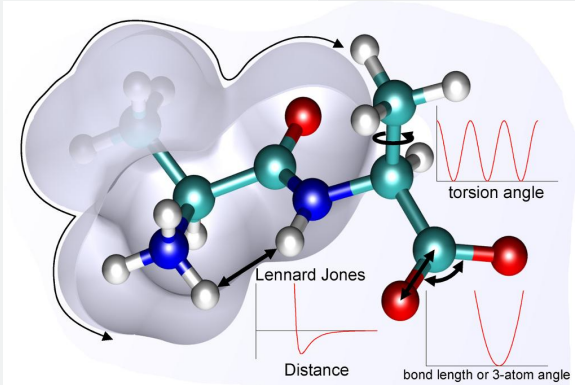


taken from Wikipedia

$$V(r_1, r_2, \ldots, r_N) = V_{bond} + V_{angle} + V_{dihedral} + V_{Coulomb} + V_{vdW}$$

Force field: basic information

- Force field – derived by fitting to experimental and/or quantum chemical data.
- Force fields are called “empirical potentials”.
  - They are not “purely” empirical, because quantum chemical data are also used for their derivation.
- There is a lot of “magic” in the force fields too.
  - It is not necessarily bad.
  - Compromises should be made.



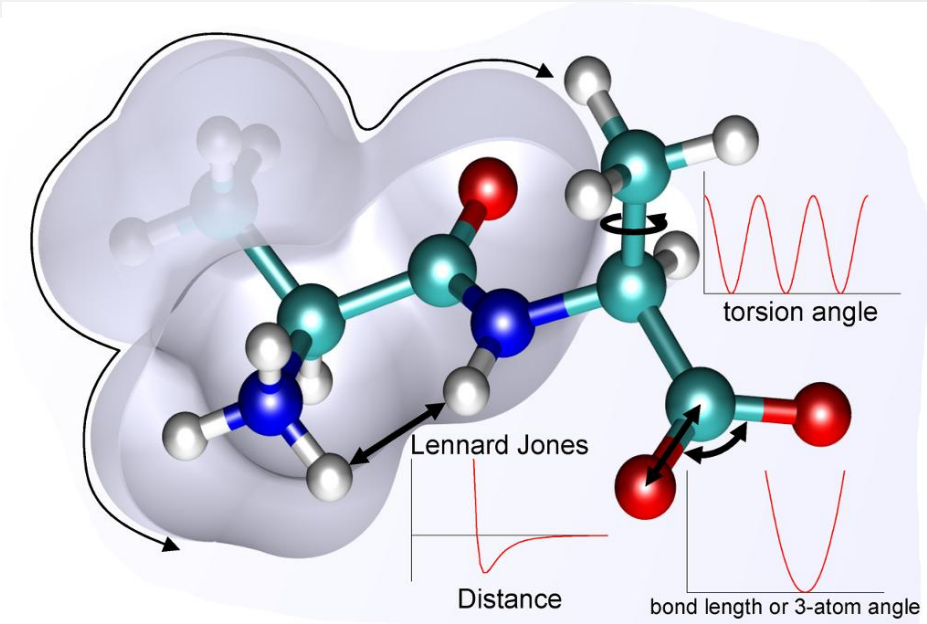
taken from Wikipedia

$$V(r_1, r_2, \ldots, r_N) = V_{bond} + V_{angle} + V_{dihedral} + V_{Coulomb} + V_{vdW}$$

Force field: basic information

- Typical biological simulation model does not have chemical reactions.

- Intramolecular interactions (chemical, bonding):
  - Interactions of atoms within one molecule.
  - Bonded interactions: bonds, angles, dihedrals.
- Intermolecular interactions (physical, non-bonding):
  - Interactions of atoms belonging to different molecules.
  - Nonbonded interactions: generally speaking, electrostatic and van der Waals (many details though).

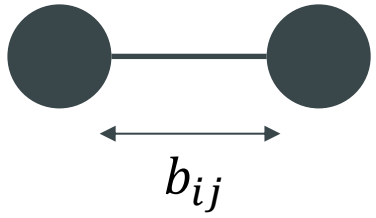


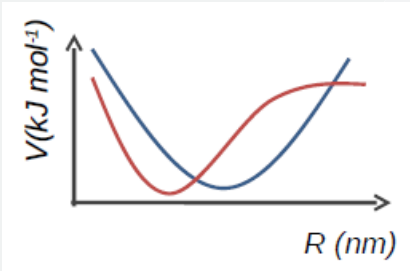
taken from Wikipedia

$$V(r_1, r_2, \ldots, r_N) = V_{bonded} + V_{nonbonded}$$

Bonded parameters: V_{bond}

- Define “bonds” (no surprise), their equilibrium length, and stiffness.
- Typically defined by either:
  - Harmonic bond potential:
    - The most popular one in bio force fields.
    - Less accurate when studying events involve significant bond stretching.
    - Computationally cheaper.
    - Still good.
  - Morse bonds potential:
    - Not currently popular.
    - More accurate when out of equilibrium.
    - More computationally expensive.
    - Good.
- Quite often, the bonds are constrained.
  - It allows larger time step.





$$V_{bond-h}(r_{ij}) = \frac{1}{2} k_{ij}^{(b)} \left(r_{ij} - b_{ij}\right)^2$$

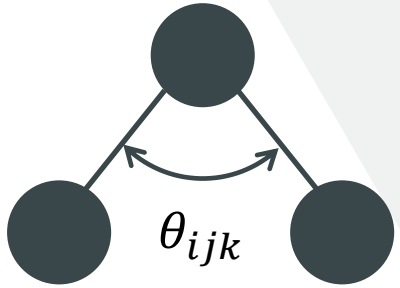
$$V_{bond-m}(r_{ij}) = D_{ij} \left[1 - \exp\left(-\beta_{ij}\left(r_{ij} - b_{ij}\right)\right)\right]^2$$

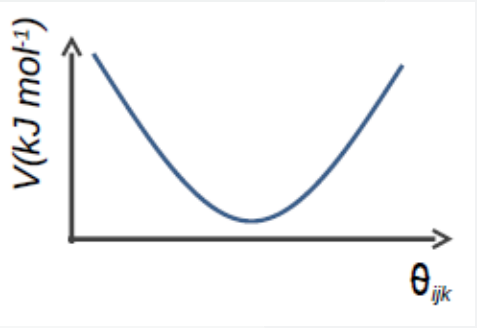
Angle parameters: V_{angle}

- Define angles, their equilibrium value, and stiffness.

- Typically defined by either:
  - Harmonic angle potential.
    - The most popular one in bio force fields.
    - Good.
  - Cosine-based potential.
  - Restricted bending potential.
  - Urey–Bradley potential.

- Angles can also be constrained in solvents.
  - Water (almost always).
  - The model being rigid implies that the geometry is fixed (both bonds and angles).

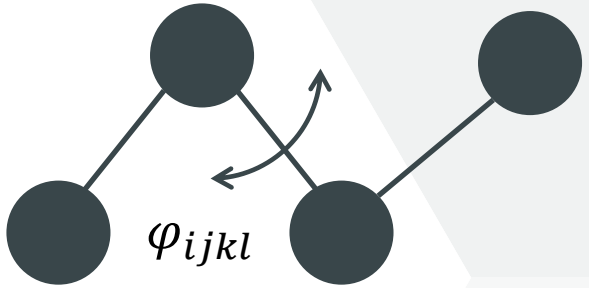


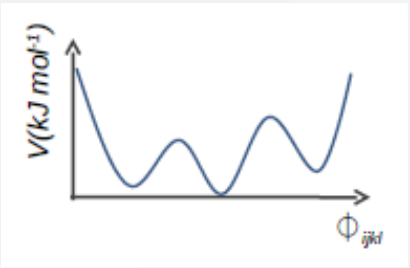


$$V_{angle-h}(\theta_{ijk}) = \frac{1}{2} k_{ijk}^{(\theta)} \left(\theta_{ijk} - \theta_{ijk}^{(0)}\right)^2$$

Dihedral (proper) parameters: $V_{dihedral}$

- Define dihedrals – torsional angles formed by four consecutively bonded atoms.
  - Angle between two consecutive planes (one defined by first three atoms, second defined by last three atoms).





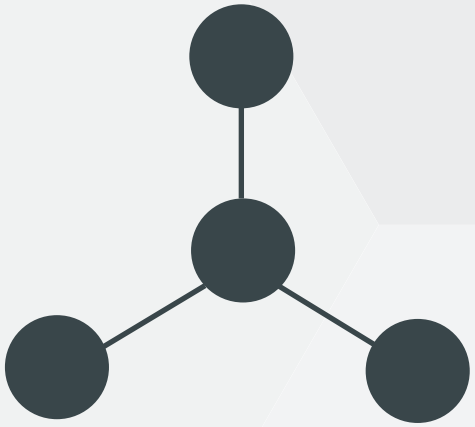
- Functional forms:
  - Periodic (Fourier, proper) (formula to the right).
  - Ryckaert–Bellemans potential.

$$V_{dihedral}(\varphi_{ijkl}) = k_{ijkl}^{(\varphi)}(1 + \cos(n\varphi_{ijkl} - \delta))$$

- Dihedrals are essential in conformation and dynamics of biomolecules.
  - Backbone rotations, nucleic acids, etc.

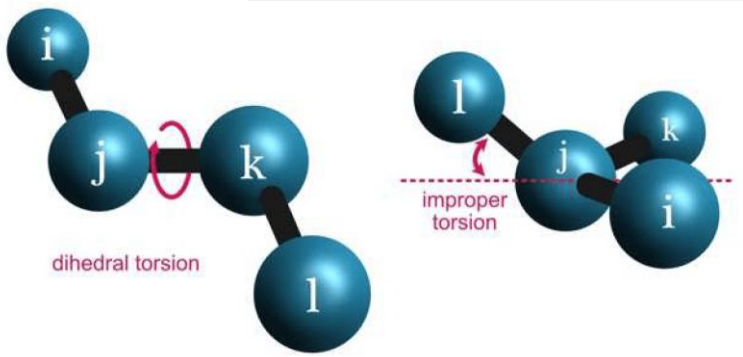
  - n – periodicity/multiplicity of the dihedral term (how many minima)
  - δ – phase shift (aka angular offset)
    - Typically, 0 for odd n and 180 for even n

Dihedral (improper) parameters: $V_{dihedral}$



- Another possible dihedral (improper one) – one atom connected to three others.

- Can be defined either as:
  - Harmonic potential (bottom–right).
    - Common choice.
    - Good.
  - Or using the same potential as for (proper) dihedral.



Fernandes et al. *Molecules* 2022, 27, 5524

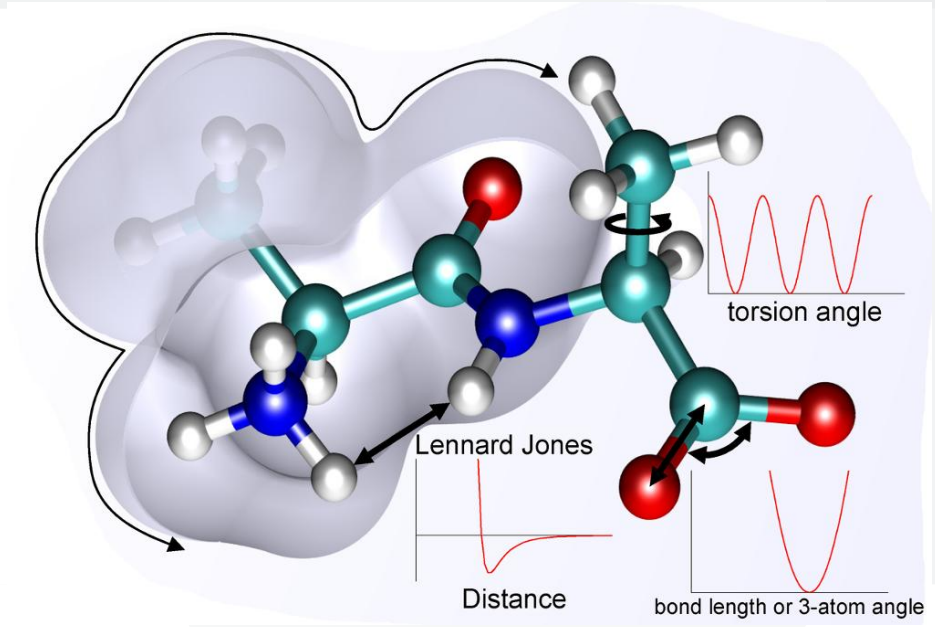
- Improper dihedrals (impropers) are also essential in conformation and dynamics of biomolecules.
  - Planar groups, chirality, etc.

$$V_{dihedral-imp}(\varphi_{ijkl}) = \frac{1}{2} k_{ijkl}^{(\varphi)} \left(\varphi_{ijkl} - \varphi_{ijkl}^{(0)}\right)^2$$

Nonbonded interactions: $V_{nonbonded}$

- Electrostatic interactions:
  - Charge–charge interactions (two ions).
  - Charge–dipole interactions (ion and water molecule).
  - Charge–induced dipole interactions (ion and non-polar atom/molecule).
  - Charge–(higher multipoles).

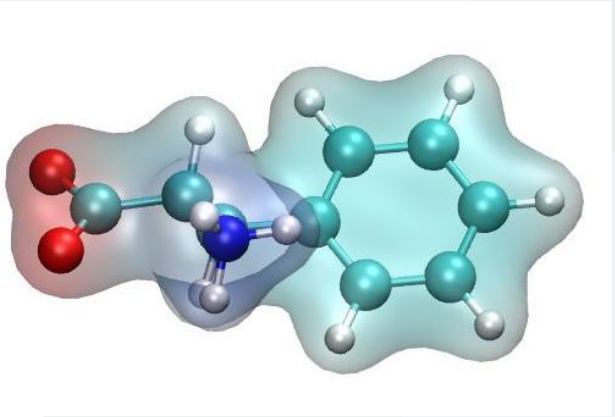
- van der Waals (vdW) interactions (all but the above, both attractive and repulsive, between uncharged particles):
  - Dipole–dipole interactions (water with another water).
  - Dipole–induced dipole, or Debye force (water with non-polar atom/molecule).
  - Induced dipole–induced dipole, or London dispersion force (two non-polar atoms/molecules).
  - Pauli repulsion (repulsive interaction, more details later).



taken from Wikipedia

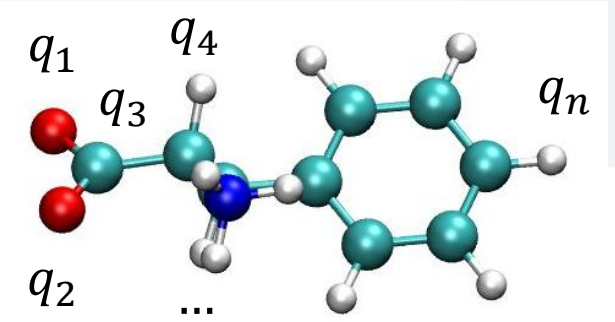
Electrostatics is the core

- All interactions at molecular level have their origin in electrostatics.



- Biosimulations typical use nonpolarizable force fields.
  - Electrostatic interactions are approximated as charge–charge interactions.

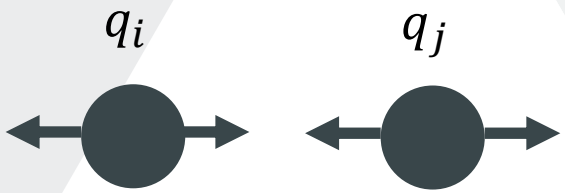
- Electronic clouds represented as point charges.
  - Derived from the approximate electrostatic potential.
  - Electrostatic potential is calculated from quantum mechanics methods exist (ESP, RESP, etc.)
  - Atomic partial charges are not experimental observables, definition is arbitrary.

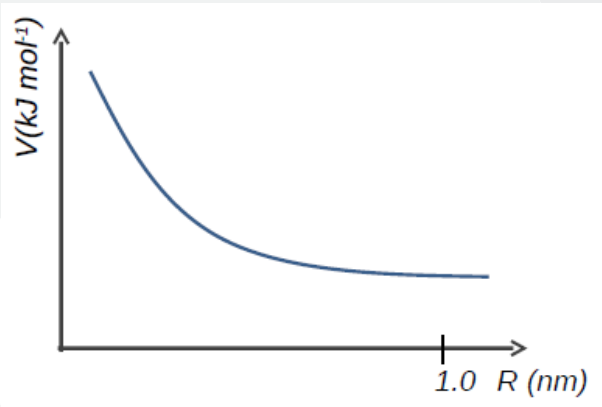


Charge–charge interactions: $V_{Coulomb}$

- Described by Coulomb law:

$$V_{Coulomb}(r_{ij}) = \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$$





- Charge–charge interactions drive multiple processes.
  - Dominant interactions.
  - Both short-range and long-range.

- Other contributors (like charge–dipole) are smaller.
  - Do not appear explicitly in nonpolarizable force fields.
  - Appear explicitly in polarizable force fields (outside the scope of the course).

van der Waals (V_{vdW}) attractive interactions

Interaction	Dependence on distance	Typical energy (kJ/mol)
charge–charge	$1/r$	~250
charge–dipole	$1/r^2$	~15
dipole–dipole	$1/r^3$	~2
dipole–dipole (rotating)	$1/r^6$	~0.6
London force	$1/r^6$	~2
hydrogen bond		~20

based on Atkins' Physical Chemistry

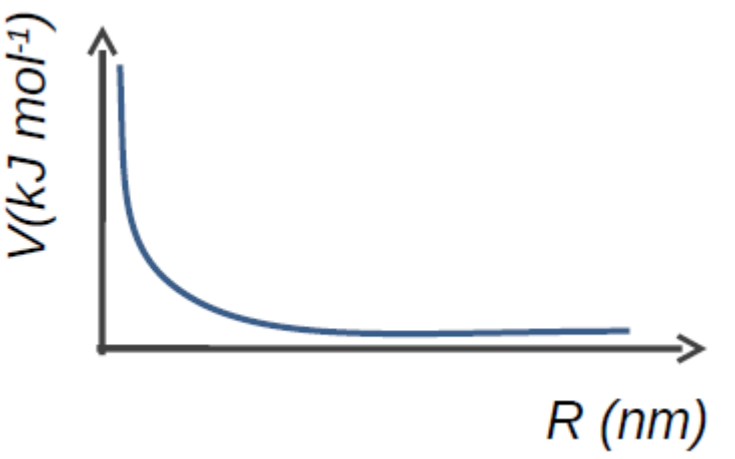


- Attractive vdW interactions can be approximated as:
  - $C_{ij}^{(6)}$ – related to the strength of attractive interactions

$$V_{vdW-att}(r_{ij}) = -\frac{C_{ij}^{(6)}}{r_{ij}^6}$$

van der Waals (V_{vdW}) repulsive interactions

- Short-range vdW interactions are strongly repulsive:
  - Pauli repulsion.
  - Overlap of electronic densities.
    - Two electrons cannot occupy the same quantum state at the same time.
  - Decays to zero at long distances.



- Originates from quantum mechanical exchange energy between fermions.

$$V_{vdW-rep}(r_{ij}) = \frac{C_{ij}^{(12)}}{r_{ij}^{12}}$$

- Can also be approximated (formula to the right)
  - $C_{ij}^{(12)}$ – related to the strength of repulsive interactions



van der Waals as Lennard–Jones: $V_{vdW} = V_{LJ}$

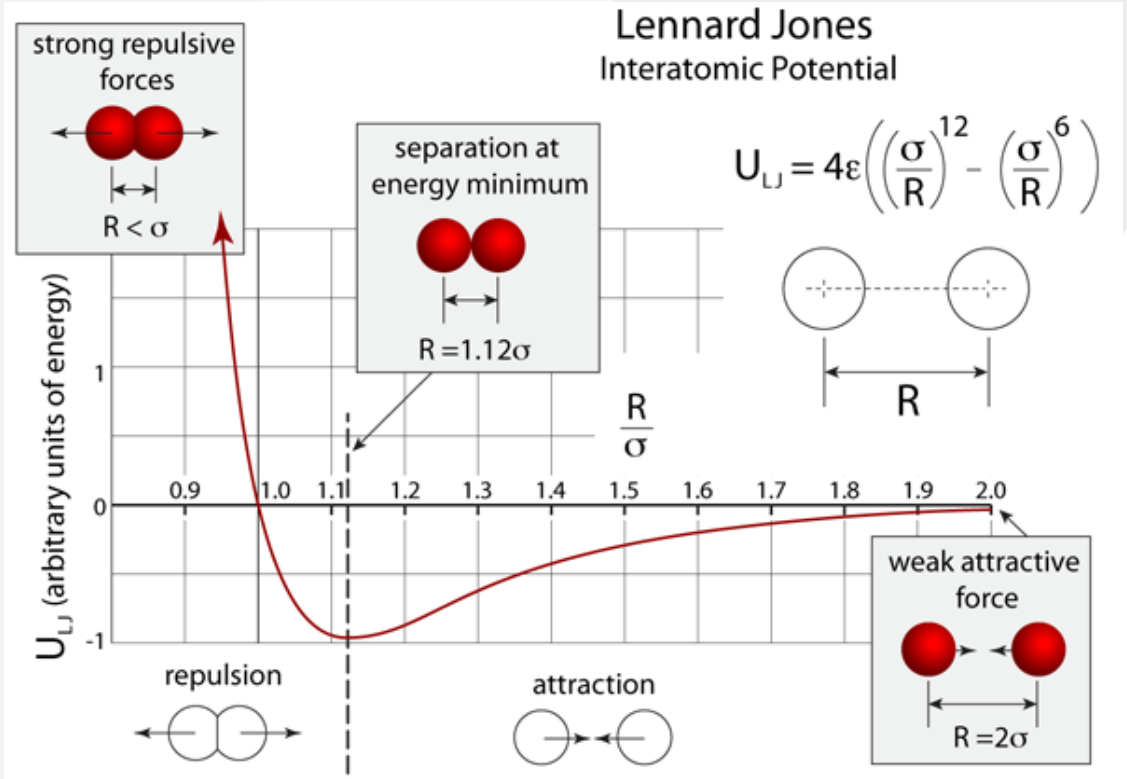
- Attractive and repulsive components constitute vdW interactions
  - C_{12} – related to the strength of repulsive interactions
  - C_6 – related to the strength of attractive interactions

$$V_{vdW}(r_{ij}) = \frac{C_{ij}^{(12)}}{r_{ij}^{12}} - \frac{C_{ij}^{(6)}}{r_{ij}^{6}}$$

- Can be approximated as Lennard–Jones potential:
  - σ_{ij} – finite distance at which the potential is zero
  - ϵ_{ij} – depth of the potential that indicate the strength of the attraction

$$V_{LJ}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$

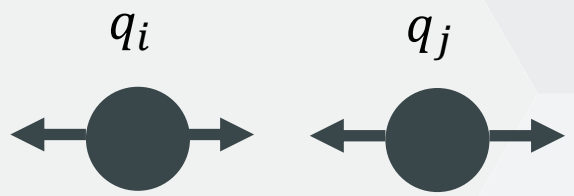
Lennard–Jones potential in a nutshell



http://atomsinmotion.com/book/chapter5/md

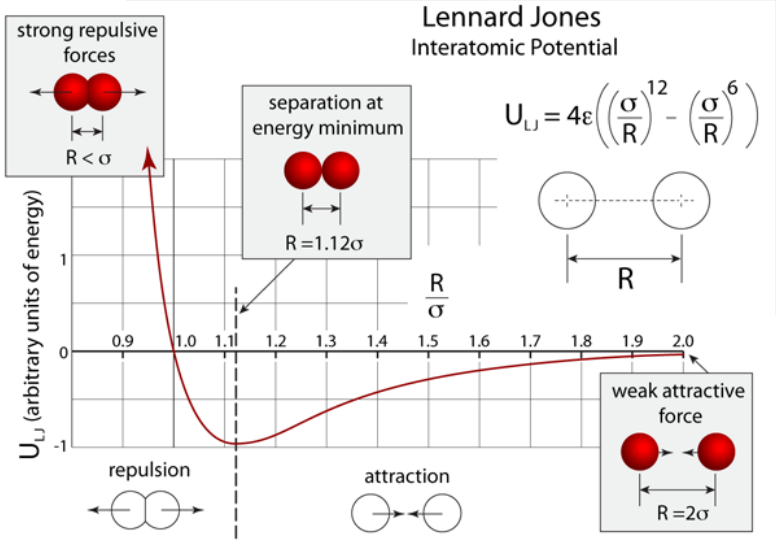
- Atoms always try to minimize their potential energy.
- At the lowest temperatures (lowest kinetic energy), they are falling to the bottom of the potential curve.
- Deviation to the left – atoms start to repel.
- Deviation to the right – atoms start to attract each other.

Nonbonded interactions: $V_{nonbonded}$



- Electrostatic interactions:
  - **Charge–charge interactions (two ions).**
  - Charge–dipole interactions (ion and water molecule).
  - Charge–induced dipole interactions (ion and non-polar atom/molecule).
  - Charge–(higher multipoles).

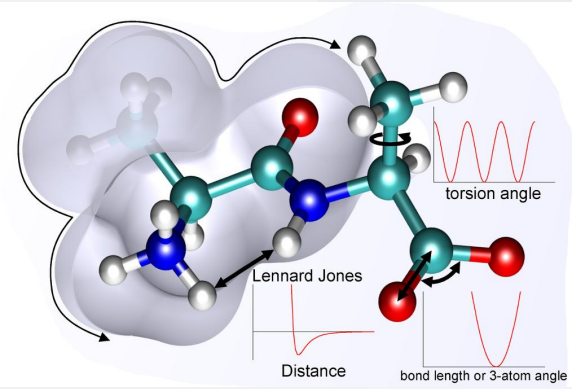
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  - Dipole–dipole interactions (water with another water).
  - Dipole–induced dipole, or Debye force (water with non-polar atom/molecule).
  - Induced dipole–induced dipole, or London dispersion force (two non-polar atoms/molecules).
    - **Simplified to attractive term.**
  - **Pauli repulsion (repulsive interaction).**



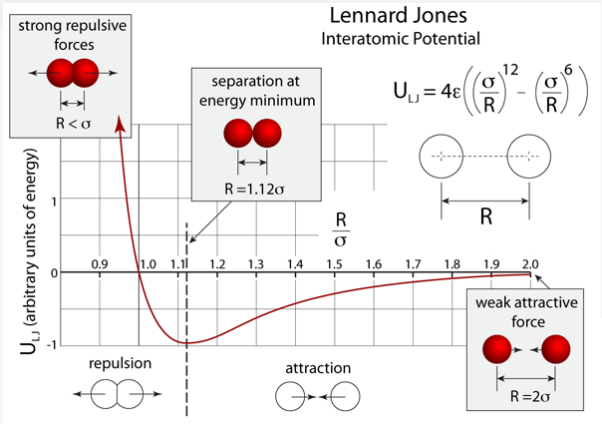
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Force fields: remarks

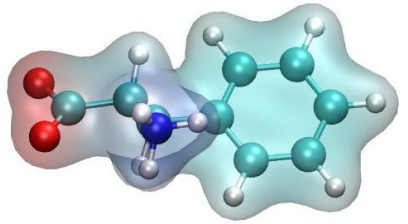
- Based on experiments, quantum chemistry data, or until it works.
- Embodied in interaction models, which are practical and fast.
- Quite often system-specific, sometimes – field-specific
  - There are force field designed for specific type of molecules (e.g., proteins).
  - There are more generic force fields, like those for ions, biomolecules, solids, etc.
- Today we discussed nonpolarizable force fields.
  - Explicitly/implicitly polarizable force fields are another level of complexity.
    - Consider not only charge–charge interactions.
    - Supposed to be better but usually not.
- Not suitable/problematic for certain phenomena (e.g., chemical reactivity, excited states) or conditions (low temperature, high pressure, unless intentionally designed).
  - Other devil details.
- **FORCE FIELDS ARE THE KEY TO MOLECULAR DYNAMICS.**
  - **Knowledge comes with experience and monitoring literature.**



taken from Wikipedia



http://atomsinmotion.com/book/chapter5/md



Cutoffs: what is that?

- A predefined distance beyond which interactions are ignored or treated approximately.
- For van der Waals interactions, beyond certain distances interactions are negligible.
  - Can be ignored.
  - Saves computational time.
- For electrostatics, cannot be ignored.
  - Long-range nature of interactions.
  - Approximated.
    - For instance, using Particle Mesh Ewald Method.
    - Saves computational time.
- In both cases, interactions are usually not suddenly truncated.
  - To not cause artifacts.

if r_{ij} < cutoff:

$$V_{LJ}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$

$$V_{Coulomb}(r_{ij}) = \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$$

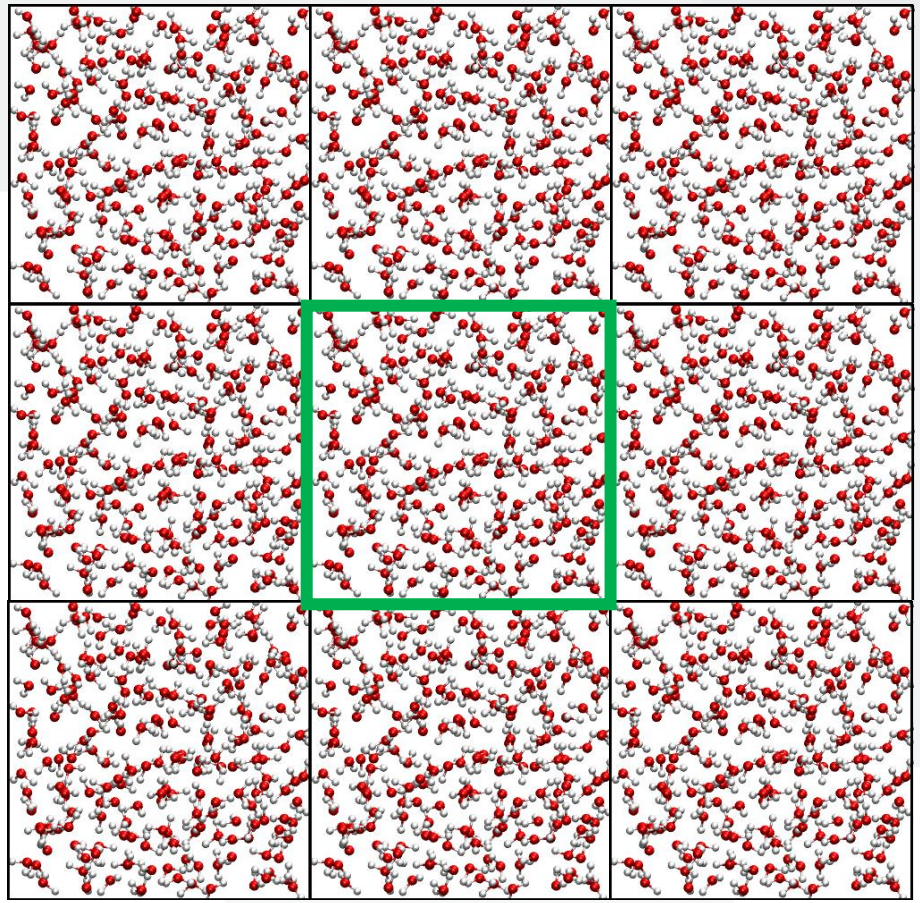
if r_{ij} > cutoff:

$$V_{LJ}(r_{ij}) = 0$$

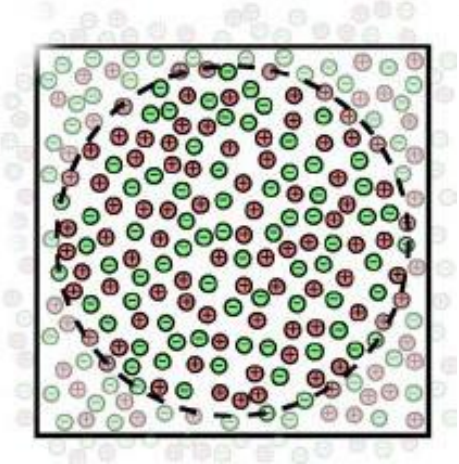
$$V_{Coulomb}(r_{ij}) = \text{approximated}$$

Particle Mesh Ewald (PME)

- A technique to efficiently compute long-range electrostatic interactions.
- PME divides electrostatic interactions into two components:
  - Real space (Coulomb law as we discussed).
  - Reciprocal space (mostly from periodic images).
    - Fast Fourier transformation.
    - Interpolation (particles charges put onto a mesh to compute the potential).



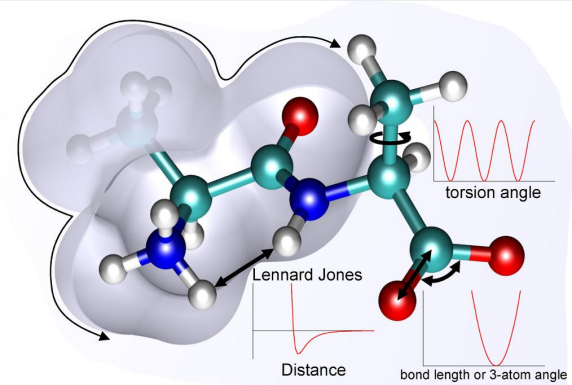
- Conditions:
  - System is in PBC.
  - Total charge of the system is zero.
- Advantages:
  - Highly accurate.
  - Flexible.



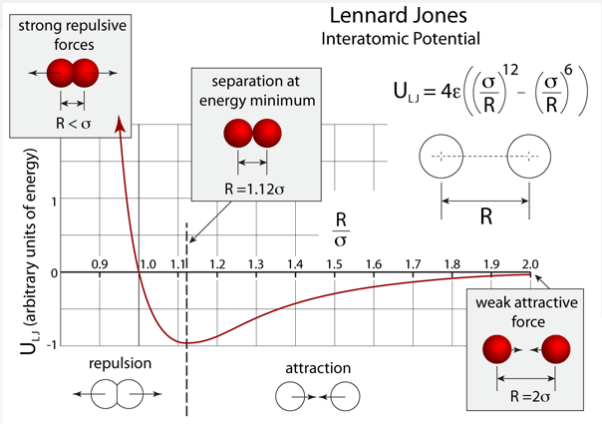
Fennell and Gezelter, *J. Chem. Phys.* 2006, 124, 234104.

Force fields: more remarks

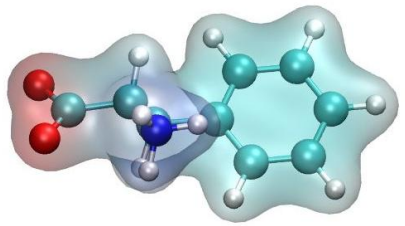
- Cutoff values are force-field specific.
  - If the force field was designed using specific cutoff, PME method or alternative, then this protocol should be used in all other simulations with this force field.
- Yes, simulations have protocols like wet-lab experiments.
- The same applies to things like time step, type of thermostat and barostat, etc.



taken from Wikipedia



http://atomsinmotion.com/book/chapter5/md



Machine learning force fields?

- Machine learning as a tool to explore parameter space
  - ML speeds up the exploration of the parameter space without altering the functional form of the force field.
  - The same principal components (partial charges, LJ parameters, etc.).
  - Increasingly common.

- Neural network force fields
  - Learn directly from quantum mechanical data.
  - Flexible functional form, no need for predefined parameters.
  - Huge potential, currently challenges with speed and reliability.



Break!

Get ready for practical session!